



SYDNEY BOYS HIGH SCHOOL
MOORE PARK, SURRY HILLS

2004

YEAR 11

HALF YEARLY EXAMINATION

Mathematics Extension

General Instructions

- Working time – 90 minutes.
- Reading Time – 5 minutes.
- Write using black or blue pen.
- Only Board approved calculators may be used.
- All necessary working should be shown in every question if full marks are to be awarded.
- Start each section in a **SEPARATE** answer booklet.
- Marks may not be awarded for messy or badly arranged work

Total Marks - 82

- Attempt Sections A – D
- All sections are **NOT** of equal value.

Examiner: *C. Kourtesis*

1.	Convert	(a)	$\frac{5\pi}{2}$ radians to degrees	1
		(b)	135° to radians	1
2.	The point $(9, k)$ lies on the line $x + 3y = 6$ Find the value of k			2
3.	If $m(x) = 3^x + 3^{-x}$ find $m(-3)$			1
4.	Solve the inequalities			
	(a)	$-4x + 1 \leq 12x$		2
	(b)	$ x < 4$		1
5.	Find the point of intersection of the straight lines $x + 2y = 6$ and $x - 3y = 10$			2
6.	On separate diagrams sketch the graphs of:			
	(a)	$xy = 4$		1
	(b)	$y = 2^{-x}$		1
	(c)	$y = \log_{10} x$		1
7.	Find the roots of the equation	$\theta^2 - \theta - 2 = 0$		2
8.	Sketch the region in the number plane indicated by $y \geq x^2$			2
9.	The equation of a parabola is given by $x^2 = 8y$ Find the	(a)	equation of the directrix	1
		(b)	coordinates of the focus	2

SECTION B (Start a NEW Booklet)
[18 marks]

10. Explain why $f(x) = \sin x$ represents an odd function 2
11. For the function $f(x) = \frac{1}{(x+1)(1-3x)}$ write down the
- (a) equations of the vertical asymptotes 2
- (b) y intercept 1
12. The equation of a circle is given by
- $$x^2 + y^2 + 4x - 8y = 0$$
- 4
- Find the
- (a) coordinates of the centre
- (b) length of the radius
13. Find the acute angle between the lines, answer to the nearest minute.
- $$y = 2x - 1 \quad \text{and} \quad y = -\frac{1}{3}x$$
- 3
14. Given the quadratic equation $2x^2 - mx + 8 = 0$
- Find the
- (a) discriminant 1
- (b) value(s) of m for which the above equation has 2
two distinct real roots
15. Find the coordinates of the point that divides the interval joining 3
 $A(1,4)$ and $B(5,10)$ externally in the ratio 3:2

SECTION C (Start a NEW Booklet)
[23 marks]

16. Solve the inequalities

(a) $(x+4)(x-2)(x-3) > 0$ 2

(b) $\frac{x-3}{1-x} > 2$ 4

17. Find the equation of the locus of a point $P(x, y)$ which moves so that its distance from $x - y + 2 = 0$ is equal to its distance from the point $(1, -1)$ 3

18. Solve the inequality $\frac{1}{|4-3x|} < 4$ 3

19. (a) Write down the algebraic definition of $|x|$ 1

(b) Sketch the graph of $y = x|x|$ 3

20. Find the value(s) of k if the equation

$$x^2 - 3kx + (k+3) = 0$$

has:

(a) one root that is double the other 2

(b) one root that is the reciprocal of the other 2

21. Solve the equation

$$1 - \cos x - 2\sin^2 x = 0 \quad \text{for } 0 \leq x \leq 2\pi$$
 3

SECTION D (Start a NEW Booklet)
[21 marks]

22. Solve $x^6 - 7x^3 - 8 = 0$ 3
23. Solve the inequality $(x + \frac{1}{x})^2 - (x + \frac{1}{x}) \leq 6$ 4
24. Prove that
- $$\frac{\sin 2\theta + \sin \theta}{\cos 2\theta + \cos \theta + 1} = \tan \theta$$
- 4
25. (a) Show that $\tan 2\theta = \frac{2 \tan \theta}{1 - \tan^2 \theta}$ 2
- (b) Hence show that
- $$\tan 15^\circ = 2 - \sqrt{3}$$
- 3
26. A man travelling along a straight flat road passes three points at intervals of 200m. From those points he observes the angle of elevation of the top of a hill to the left of the road to be respectively 30° , 45° and again 45° .
- (a) Draw a neat diagram to represent this information 1
- (b) Find the height of the hill 4

THIS IS THE END OF THE EXAMINATION



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Sample Solutions

Section	Marker
A	Mr Choy
B	Mr Parker
C	Mr Hespe
D	Mr Bigelow

Section A

(1) (a) $\pi^c = 180$
 $\frac{5\pi}{2} = 180 \times \frac{5}{2} = 450$

(b) $180 = \pi^c$
 $\frac{35}{180} \times 180 = \pi \times \frac{35}{180} = \frac{3\pi}{4}$

(2) $9 + 3k = 6$
 $3k = -3$
 $k = -1$

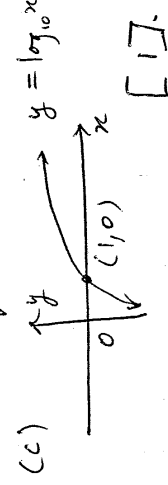
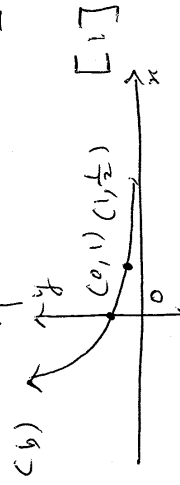
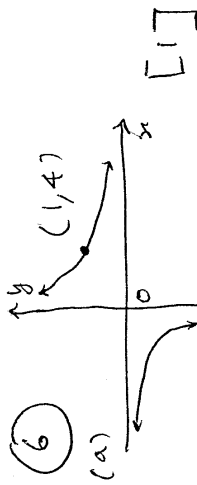
(3) $m(-3) = 3^{-3} + 3^3$
 $= 27\frac{1}{27}$
 (27.04)

(4) (a) $-4x + 1 \leq 12x$
 $16x \geq 1$
 $x \geq \frac{1}{16}$

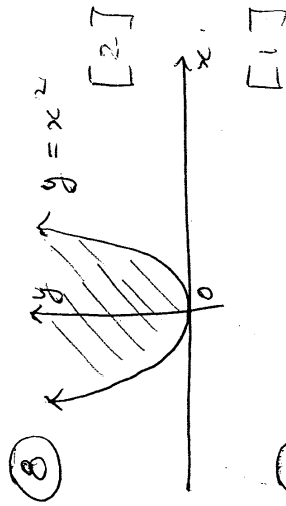
(b) $|x| < 4$
 $-4 < x < 4$

(5) $x + 2y = 6$
 $x - 3y = 10$
 $-5y = -4$
 $y = \frac{4}{5}$

$x - 8/5 = 6$
 $x = 7\frac{3}{5} = (38/5)$
 $\therefore$ pt of intersection
 $(7\frac{3}{5}, -4/5)$



(7) $\theta^2 - \theta - 2 = 0$
 $(\theta - 2)(\theta + 1) = 0$
 $\therefore \theta = -1, 2$



(9) $x^2 = 8y$, $4a = 8$
 $a = 2$

(a) $y = -2$ equation of directrix
 (b) (0, 2) focus

SECTION B [18 marks]

10. An odd function satisfies $f(-x) = -f(x)$

$$\therefore f(-x) = \sin(-x) = -\sin x = -f(x)$$

11.
$$f(x) = \frac{1}{(x+1)(1-3x)}$$

- (a) The equations of the vertical asymptotes are $x = -1$ and $x = \frac{1}{3}$.

(b) y intercept is when $x = 0 \Rightarrow f(0) = \frac{1}{(0+1)(1-0)} = 1$.

So the y - intercept is $(0,1)$.

12. $x^2 + y^2 + 4x - 8y = 0$

$$\therefore (x^2 + 4x) + (y^2 - 8y) = 0$$

$$\Rightarrow \left(x^2 + 4x + \left(\frac{4}{2} \right)^2 \right) + \left(y^2 - 8y + \left(\frac{-8}{2} \right)^2 \right) = 0 + 4 + 16$$

$$\Rightarrow (x^2 + 4x + 4) + (y^2 - 8y + 16) = 20$$

$$\Rightarrow (x+2)^2 + (y-4)^2 = 20$$

- (a) coordinates of the centre are $(-2,4)$

- (b) length of the radius is $\sqrt{20} = 2\sqrt{5}$

13. The acute angle between the lines is given by $\tan \theta = \left| \frac{m_1 - m_2}{1 + m_1 m_2} \right|$.

$$y = 2x - 1 \Rightarrow m_1 = 2 \text{ and } y = -\frac{1}{3}x \Rightarrow m_2 = -\frac{1}{3}$$

$$\tan \theta = \left| \frac{2 - (-1/3)}{1 + (2)(-1/3)} \right| = 7 \Rightarrow \theta = 81^\circ 52'$$

14. $2x^2 - mx + 8 = 0$

(a) The discriminant is $\Delta = b^2 - 4ac = (-m)^2 - 4 \times 2 \times 8 = m^2 - 64$

(b) Two distinct real roots $\Rightarrow \Delta > 0$ ie $m^2 - 64 > 0$

$$\Rightarrow (m-8)(m+8) > 0$$

$$\therefore m < -8, m > 8$$

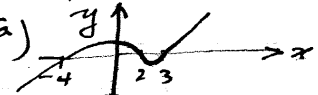
15. $P\left(\frac{mx_2 + nx_1}{m+n}, \frac{my_2 + ny_1}{m+n}\right)$ where $A(1,4) = (x_1, y_1)$, $B(5,10) = (x_2, y_2)$ and

$$m : n = 3 : -2.$$

$$P\left(\frac{3 \times 5 + (-2) \times 1}{3-2}, \frac{3 \times 10 + (-2) \times 4}{3-2}\right) = (13, 22)$$

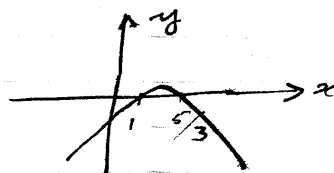
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Part C

16: (a)  $-4 < x < 2, x > 3$

(b) Method 1:

$$\begin{aligned}(x-3)(1-x) &> 2(1-x)^2 \\ (1-x)(x-3-2[1-x]) &> 0 \\ (1-x)(3x-5) &> 0 \\ 1 < x < 5/3\end{aligned}$$



Method 2:

$$\begin{aligned}\text{If } x > 1, \quad x-3 &< 2(1-x) \\ 3x &< 5 \\ x &< 5/3.\end{aligned}$$

$$\begin{aligned}\text{If } x < 1, \quad x-3 &> 2(1-x) \\ 3x &> 5 \\ x &> 5/3\end{aligned}$$

Contradiction.

$$\therefore 1 < x < 5/3.$$

17 $\frac{|x-y+2|}{\sqrt{2}} = \sqrt{(x-1)^2 + (y+1)^2}$

$$\frac{x^2 + y^2 + 4 - 2xy + 4x - 4y}{2} = x^2 - 2x + 1 + y^2 + 2y + 1$$

$$\begin{aligned}x^2 + y^2 + 4 - 2xy + 4x - 4y &= 2x^2 - 4x + 4 + 2y^2 + 4y \\ x^2 + y^2 + 2xy - 8x + 8y &= 0\end{aligned}$$

18 $1 < 4|4-3x|$

Method 1:

$$\begin{aligned}\text{If } x < 4/3 \\ 1 &< 16 - 12x\end{aligned}$$

$$12x < 15$$

$$x < 5/4$$

$$\text{If } x > 4/3$$

$$1 < 12x - 16$$

$$12x > 17$$

$$x > 17/12$$

As $5/4 < 4/3$ and $17/12 > 4/3$
 then $x < 5/4$ and $x > 17/12$

Method 2:

$$1 < 16(16 - 24x + 9x^2)$$

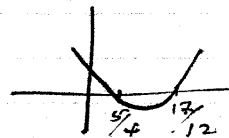
$$1 < 256 - 384x + 144x^2$$

$$0 < 255 - 384x + 144x^2$$

$$0 < 85 - 128x + 48x^2$$

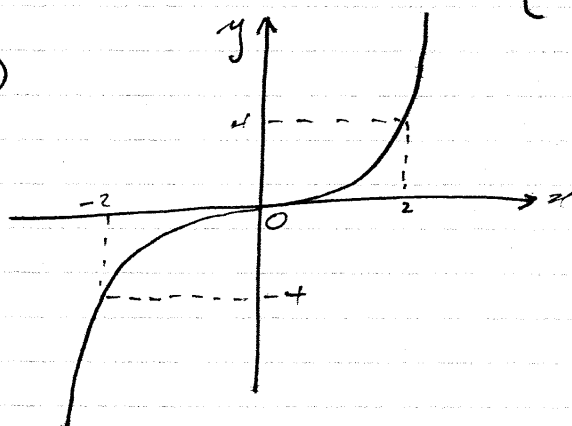
$$0 < (12x - 17)(4x - 5)$$

$$\therefore x < 5/4 \text{ and } x > 17/12$$



19 (a) $|x| = \sqrt{x^2}$ or $|x| = \begin{cases} x & \text{if } x \geq 0 \\ -x & \text{if } x < 0 \end{cases}$

(b)



$$y = x^2 \text{ when } x \geq 0 \\ y = -x^2 \text{ when } x < 0$$

20 Not counted (a) $k = -1, 3/2$
 (b) $k = -2$

$$21 \quad 1 - \cos x - 2(1 - \cos^2 x) = 0$$

$$2\cos^2 x - \cos x - 1 = 0$$

$$(2\cos x + 1)(\cos x - 1) = 0$$

$$\cos x = -1/2, 1$$

$$x = 2\pi/3, 4\pi/3, 0, 2\pi$$

SECTION D.

$$22. x^6 - 7x^3 - 8 = 0$$

$$(x^3 - 8)(x^3 + 1) = 0$$

$$x^3 = 8, -1$$

$$\boxed{x = 2, -1}$$

$$23. \left(x + \frac{1}{x}\right)^2 - \left(x + \frac{1}{x}\right) \leq 6.$$

$$\left(x + \frac{1}{x}\right)^2 - \left(x + \frac{1}{x}\right) - 6 \leq 0$$

$$\left(x + \frac{1}{x} - 3\right)\left(x + \frac{1}{x} + 2\right) \leq 0$$

$$\therefore -2 \leq x + \frac{1}{x} \leq 3.$$

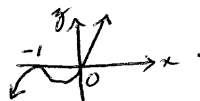
Consider $x + \frac{1}{x} \geq -2$.

$$x + \frac{1}{x} + 2 \geq 0$$

$$\frac{x^2 + 1 + 2x}{x} \geq 0 \quad (\text{Multiply both sides by } x^2)$$

$$x(x^2 + 2x + 1) \geq 0$$

$$x(x+1)^2 \geq 0$$



$$\therefore x = -1, x \geq 0 \quad \text{NB } x \neq 0$$

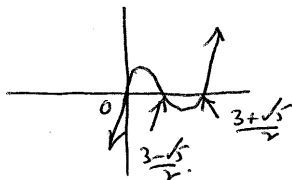
$$\therefore \boxed{x = -1, x \geq 0} \quad (A)$$

Consider $x + \frac{1}{x} \leq 3$

$$x^2 - 3x + 1 \leq 0 \quad (\text{Multiply both sides by } x^2)$$

$$x(x^2 - 3x + 1) \leq 0 \quad (\text{Now solving } x^2 - 3x + 1 = 0)$$

$$x = \frac{3 \pm \sqrt{5}}{2}$$



$$\therefore \boxed{\frac{3-\sqrt{5}}{2} \leq x \leq \frac{3+\sqrt{5}}{2}, x < 0} \quad (B)$$

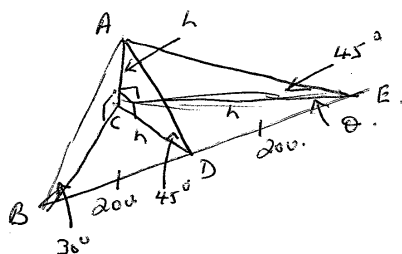
Taking the intersection of sets (A) & (B)

$$\boxed{x = -1, \frac{3-\sqrt{5}}{2} \leq x \leq \frac{3+\sqrt{5}}{2}}$$

$$\begin{aligned} 24. \quad \underline{\hspace{1cm}} \quad LHS &= \frac{\sin 2\theta + \sin \theta}{\cos 2\theta + \cos \theta + 1} \\ &= \frac{2\sin \theta \cos \theta + \sin \theta}{2\cos^2 \theta - 1 + \cos \theta + 1} \\ &= \frac{2\sin \theta \cos \theta + \sin \theta}{2\cos^2 \theta + \cos \theta} \\ &= \frac{\cancel{\sin \theta} (2\cos \theta + 1)}{\cos \theta (2\cos \theta + 1)} \\ &= \frac{\cancel{\sin \theta}}{\cancel{\cos \theta}} \\ &= \tan \theta \\ &= RHS. \end{aligned}$$

26.

(a)



NB Δ 's ACD and ACE are isosceles
& right angled

Let $AC = CD = CE = h$.

$q_m \supset A \subset B, \quad BC = \sqrt{3}h.$

Let $LCB = \emptyset$.

using Cosine Rule twice.

$$C_{100} = \frac{400^2 + x^2 - 3x^2}{2 \times 400 \times x} \quad (A)$$

$$\therefore \text{LSD} = \frac{200^2 + x^2 - x^2}{2 \times 200 \times x} \quad (B)$$

25(a) Manig

$$\tan(\alpha + \beta) = \frac{\tan \alpha + \tan \beta}{1 - \tan \alpha \tan \beta}$$

Let $\alpha = \beta = 0$

$$\therefore \tan(\theta + \theta) = \frac{\tan \theta + \tan \theta}{1 - \tan \theta \tan \theta}$$

$$\therefore \boxed{\tan 2\theta = \frac{2 \tan \theta}{1 - \tan^2 \theta}} \quad (A)$$

(b) NO question states Hence!!

Iron (A)

$$\tan 30^\circ = \frac{2 \tan 15^\circ}{1 - \tan^2 15^\circ}$$

Let $t = \tan 15^\circ$.

$$\therefore \frac{i}{\sqrt{3}} = \frac{2t}{1-t^2}$$

$$1 - \tau^2 = 2\sqrt{3}t.$$

$$t^2 + 2\sqrt{3}t - 1 = 0$$

$$t = \frac{-2\sqrt{3} \pm \sqrt{12+4}}{2}$$

$$\frac{-2 \pm \sqrt{32}}{2}$$

$$= -\sqrt{3} \pm 2.$$

new $t = \tan 15^\circ > 0$.

$$\therefore \tan 15^\circ = 2 - \sqrt{3}$$

$$\text{Index}(A) \times (B)$$

$$\frac{400^2 - 2x^2}{800x} = \frac{200^2}{400x}$$

$$\therefore 400^2 - 2 \cdot 200^2 = 2x^2$$

$$y^2 = 40000$$

$$x = 200 \text{ m}$$